0:00

0 seconds

All right, last one is magnesium.

0:06

6 seconds

Okay, magnesium. There we go. Magnesium we can focus on.

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15 seconds

Well, first your normal range 1.5 to 2.5. Very narrow range. Okay.

0:21

21 seconds

um functions, main functions, nerve, muscle, blood pressure, kind of the things it acts on.

0:32

32 seconds

Okay, so we got hypomagnesmia less than 1.5 and hypermagnesmia greater than 2.5.

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41 seconds

Okay, so what are the things that are going to cause hypomagnesmia?

0:47

47 seconds

Okay. Um, there's this relationship with potassium and mag. So, if I don't have mag, I can't have potassium. If I don't have potassium, I don't have the mag.

0:57

57 seconds

Okay.

1:00

1 minute

You know what's really great about hypomagnesmia?

1:03

1 minute, 3 seconds

Even though there's this relationship with potassium, magnesium is like so weird. Okay. is that um

1:12

1 minute, 12 seconds

your signs and symptoms before I get even get to the causes your signs and symptoms are those of hypocalcemia.

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1 minute, 20 seconds

So hypomagnesmia and hypocalcemia look like Kramer from Seinfeld. Okay.

1:29

1 minute, 29 seconds

Shiovastexcos spasms muscle um tetany twitching cramps

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1 minute, 36 seconds

same thing. Okay, so signs and symptoms are easy. What the heck is causing a low mag? Well, a lot

1:45

1 minute, 45 seconds

of diuretics can cause low mag. Your malnutrition.

1:52

1 minute, 52 seconds

This is the more common electrolyte imbalance that I see in your alcoholic patients um because they're so malnourished that they're very low on their magnesium.

2:04

2 minutes, 4 seconds

high PPI use, obviously a decrease intake or a decreased absorption of magnesium, okay?

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2 minutes, 12 seconds

Or a lot of excretion of magnesium.

2:16

2 minutes, 16 seconds

Um, and then your low potassium, okay.

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2 minutes, 23 seconds

So, how are we treating the low mag?

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2 minutes, 26 seconds

Well, we replace the mag either PO or IV. Okay. PO mag

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2 minutes, 33 seconds

supplements. Perfect. I could encourage magnesium rich foods, spinach,

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2 minutes, 41 seconds

pumpkin seeds, black beans, dark chocolate, and nuts.

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2 minutes, 51 seconds

Okay.

2:53

2 minutes, 53 seconds

Or I can give some IV mag. And IV mag um very high risk to give.

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3 minutes

Um, if you push it, it's got to be really, really, really freaking slow. Um, I hate that mag can be given push.

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3 minutes, 12 seconds

Whatever. Okay.

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3 minutes, 15 seconds

I'd rather just hang it. Everything I feel like I used to hang and then in Missouri it's like what are we doing?

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3 minutes, 22 seconds

Like I'm pushing mag. My gosh. So scary.

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3 minutes, 25 seconds

Anyways, um, mag, double verify the mag with another nurse, okay? Double verify

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3 minutes, 33 seconds

the dose. And you're going to push that really, really slowly because the main risk for IV mag is respiratory depression.

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3 minutes, 42 seconds

Okay.

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3 minutes, 45 seconds

If too fast, push. Okay.

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3 minutes, 52 seconds

So our nursing things um basically the same as our hypocalcemia,

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4 minutes

seizure precautions, deep tendon reflexes, cardiac monitor, neuromuscular assessment.

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4 minutes, 9 seconds

Okay, very redundant.

4:12

4 minutes, 12 seconds

Hypomagnesmia, think hypocalcemia, same thing. Okay.

4:18

4 minutes, 18 seconds

All right. hyper mag.

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4 minutes, 24 seconds

Now we got too much mag. How are we getting too much mag? Well, um, magnesium normally is excreted through our kidneys. So, if we have CKD, that's

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4 minutes, 33 seconds

going to cause high magnesium because we can't pee anything out. we have too high of intake

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4 minutes, 41 seconds

um of magnesium either sorry high magnesium intake and usually this

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4 minutes, 49 seconds

is from like a decreased gastric or um intestinal motility like decreased paristalsis which causes more things um

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4 minutes, 58 seconds

to be pulled out of the intestines like magnesium, laxatives

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5 minutes, 7 seconds

or ant acids with magnesium in it. So, a lot of laxative antacid use, trauma, again, think kidney. All right.

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5 minutes, 20 seconds

And hypothyroidism.

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5 minutes, 25 seconds

Okay. I'm not like well-versed in the thyroid part. Like, I can't really speak to the patho of that honestly.

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5 minutes, 33 seconds

Um, but yes, the rest that I could just trying to be real with you guys.

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5 minutes, 40 seconds

Okay, it's the one weird thing.

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5 minutes, 44 seconds

All right. Your signs and symptoms for hyper magnet. Oh, you know what? Um I don't know if there's well I mean

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5 minutes, 53 seconds

there's a fair number of you I feel like that are interested in OB and um postpartum and stuff but our

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6 minutes

pre-acclampsia patients and our clampic patients or patients with a clampsia

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6 minutes, 8 seconds

um the treatment for that is magnesium IV magnesium okay so um those patients if we're giving

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6 minutes, 17 seconds

them too much bag can cause a hypermagnesmia.

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6 minutes, 21 seconds

So, these are the signs and symptoms that we need to look out for. Again, confusion big. Okay, I'm actually going to write these in order. The first thing

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6 minutes, 30 seconds

that we're probably going to see in this patient is decreased deep tendon reflexes.

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6 minutes, 35 seconds

Okay, then we're going to see some hypotension develop, some flushing,

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6 minutes, 42 seconds

maybe some blurry vision, headache. Then we're going to notice a decreased respiratory rate.

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6 minutes, 48 seconds

and then they can quickly progress to cardiac arrest and they're also at risk for seizures. Okay,

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6 minutes, 56 seconds

so the treatment for this think kind of like potassium the treatment oddly

7:02

7 minutes, 2 seconds

enough is IV calcium gluconate. Okay, it's the same idea. This can quickly progress to cardiac arrest.

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7 minutes, 11 seconds

So the calcium gluca gluconate helps to um stabilize that cardiac membrane and

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7 minutes, 19 seconds

make it less likely for these dysriythmias to occur. Okay.

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7 minutes, 24 seconds

Another thing I can do are diuretics.

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7 minutes, 29 seconds

Right here we pee out potassium. So if we have too much diuretics it can cause hypo mag.

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7 minutes, 36 seconds

But it can also be used to bring down our hyper mag patients.

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7 minutes, 41 seconds

So loop diuretics are going to help. And this is another instance where like we can give them fluids, IV fluids to kind of help flush that out. If it's bad

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7 minutes, 50 seconds

enough and it's a CKD, we're doing dialysis.

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7 minutes, 53 seconds

Okay, your nursing stuff is pretty much the same. Okay, you're

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8 minutes, 1 second

assessing deep tendon reflexes. You're putting this patient on a cardiac monitor. You're monitoring their respiratory status. Um, you're

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8 minutes, 9 seconds

monitoring their neuro status, seizure precautions. Okay. And then

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we're keeping calcium at the bedside for these patients with hyper mag just

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8 minutes, 22 seconds

in case. I honestly've never seen that done, but it is nursing school. So, um,

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8 minutes, 30 seconds

it's mentioned. I feel like I should mention it anyways. Yeah. Never seen that done, but it's the same idea of the

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8 minutes, 37 seconds

um hyperc calmia. We give them calcium to stabilize that cardiac membrane. But that's mag. Okay. Let me know if you have questions.